Christian Gagné, Ph.D., ing.

Director of the Institute Intelligence and Data (IID)
Canada-CIFAR AI Chair, Associate Academic Member of Mila
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Research Orientations

My research concerns methods for deep learning and stochastic optimization, and their application to concrete problems. I work primarily on the robustness and generalization of deep neural networks under distribution shift, on neuro-symbolic approaches that make them more interpretable, and on multimodal foundation models. A significant part of this work is conducted in contact with application domains — computer vision, super-resolution microscopy, health, transportation and energy — where these methods are put to the test.

Training

- **Ph.D. in Electrical Engineering**, Université Laval, 2005.

  Thesis: *Algorithmes évolutionnaires appliqués à la reconnaissance des formes et à la conception optique*.
  Committee: Marc Parizeau (supervisor), Denis Laurendeau, Robert Sabourin, and Marc Schoenauer.
- **B.Eng. in Computer Engineering**, Université Laval, 2000.

Professional Experience

- **Director**, Institute Intelligence and Data, Université Laval, since 2019 (on sabbatical leave, January–August 2023).
- **Professor**, Electrical Engineering and Computer Engineering Department, Université Laval

  Full Professor since 2018; Associate Professor 2013–2018; Assistant Professor 2008–2013.
- **Deputy Director**, Big Data Research Centre, Université Laval, 2018–2019.
- **Research Analyst**, Research and Development Department, MacDonald, Dettwiler and Associates Ltd. (Vancouver, BC, Canada), 2007–2008.
- **Consultant**, Informatique WGZ Inc. (Québec, QC, Canada), 2006–2007.
- **Postdoctoral Fellow**, Information Systems Institute, University of Lausanne (Switzerland), 2006.
- **Postdoctoral Fellow**, TAO Team, INRIA Saclay–Île-de-France (Orsay, France), 2005–2006.
- **Lecturer**, Computer Science and Software Engineering Department, Université Laval, 2005.
- **Unix/Linux Systems Administrator**, Computer Vision and Systems Laboratory, Université Laval, 2001–2004.
- **Teaching Assistant**, Electrical Engineering and Computer Engineering Department, Université Laval, 2000–2003.
- **Consultant**, Red Queen Capital Management Inc. (Dallas, TX, USA), 2003.
- **Research Assistant**, Computer Vision and Systems Laboratory, Université Laval, 1998–2000.

Professional Affiliations

- Full professor at the Electrical Engineering and Computer Engineering Department of Université Laval
- Director of the Institute Intelligence and Data (IID) of Université Laval
- Canada-CIFAR AI Chair
- Associate Academic Member of Mila
- Member of the Computer Vision and Systems Laboratory (CVSL)
- Member of the Centre de recherche en Robotique, Vision et Intelligence Machine (CeRVIM) of Université Laval
- Board member of the Big Data Research Centre (BDRC) of Université Laval
- Member of the REPARTI (cyber-physical systems) strategic cluster of the FRQNT
- Member of the UNIQUE (neuroscience and AI) strategic cluster of the FRQNT
- Member of the CERVO (neuroscience) research centre of the FRQS
- Member of the VITAM (sustainable health) research centre of the FRQS
- Researcher-member of the Observatoire international sur les impacts sociétaux de l'IA et du numérique (OBVIA)

Public and Community Service

- Independent member of the board of directors, Musée national de l'histoire du Québec (NAT; National Museum of Québec History), since 2026.

MEDIA COVERAGE

- Print interview, "L'IA est-elle en train de devenir 'consciente'?", *Le Soleil*, November 2026.
- Print interview, "Audrey Durand, figure de proue de l'IA au Québec", *Québec Science*, May 2026.
- Radio interview, "Gemini Spark : l'avenir de la recherche web ?", *C'est encore mieux l'après-midi* (ICI Radio-Canada Première Québec), May 2026.
- Print interview, "Fabriquer l'IA québécoise sans casser des œufs", *Le Devoir*, May 2026.
- Print interview, "Protéger la diversité des langues face à l'IA", *ICI Radio-Canada Info*, April 2026.
- Print interview, "L'Institut intelligence et données de l'Université Laval reçoit 2 M$ pour consolider la formation d'un pôle d'expertise régional en intelligence artificielle", *ULaval Nouvelles*, April 2026.
- Print interview, "L'IA et la démocratie – Un faux GND pour 15 $ – C'est le temps d'être courageux", *La Presse*, March 2026.
- Podcast interview, "L'IA au service de la science, de la société et du Québec avec Christian Gagné", *IA Québec: Le balado*, January 2026.
- Podcast interview, "De chercheurs à rockstars de l'IA avec Hugo Larochelle, Aaron Courville et Christian Gagné", *IA Québec: Le balado*, December 2025.
- Co-signed opinion piece, "Plaidoyer pour l'adoption d'un code de conduite sur l'IA", *La Presse* (Opinions), October 2025.
- Co-signed opinion piece, "Il faut agir pour protéger nos démocraties", *La Presse* (Opinions), April 2025.
- Radio interview, "DeepSeek, l'agent conversationnel qui bouleverse la course mondiale à l'IA", *L'heure de pointe - Acadie* (ICI Radio-Canada Première Acadie), January 2025.
- Radio interview, "C'est quoi DeepSeek, le « ChatGPT chinois »?", *C'est encore mieux l'après-midi* (ICI Radio-Canada Première Québec), January 2025.
- Radio interview, "Retour sur la remise du prix Nobel en physique", *Le matin du Nord* (ICI Radio-Canada Première Nord de l'Ontario), October 2024.

- Radio interview, “Geoffrey Hinton de l’Université de Toronto colauréat du Prix Nobel de physique”, *Dans la mosaïque* (ICI Radio-Canada Première Toronto), October 2024.
- Podcast interview, “Une machine qui "apprend" avec peu - Données massives VS données frugales en IA - Entrevue avec Christian Gagné”, *I.A. Café*, October 2024.
- Print interview, “L’intelligence artificielle au service de la prévention du suicide”, *Le Soleil*, September 2024.
- Radio interview, “Gemini, IA de Google : quel rôle jouera-t-elle?”, *C’est encore mieux l’après-midi* (ICI Radio-Canada Première Québec), December 2023.
- Print interview, “ChatGPT, l’intelligence artificielle qui discute comme un humain”, *Urbania*, December 2022.
- Print interview, “L’intelligence humaine derrière l’IA”, *La Presse*, November 2022.
- Radio interview, “Préparer les réseaux pour l’arrivée massive des voitures électriques”, *Les années lumière* (ICI Radio-Canada Première), November 2022.
- Print interview, “Exploring the Great Unknown”, *CIFAR News*, February 2022.
- Print interview, “Le réseau d’Hydro-Québec est-il suffisant pour charger tous les véhicules électriques?”, *ICI Radio-Canada Info*, January 2022.
- Print interview, “Est-ce que l’IA fonctionnera un jour avec aussi peu d’énergie qu’un cerveau humain?”, *Le Devoir*, October 2018.
- Print interview, “Une ébullition qui va bien au-delà de Montréal”, *Le Devoir*, September 2018.
- Television interview, “Les défis réels de l’intelligence artificielle”, *Le Téléjournal Québec* (ICI Radio-Canada Télé Québec), April 2018.
- Radio interview, “Zéro émission : le pari de l’auto-Québec”, *Les années lumière* (ICI Radio-Canada Première), October 2016.
- Radio interview, “Présent et avenir des transports électrifiés, intelligents”, *Radio-Canada cet après-midi* (ICI Radio-Canada Première Québec), December 2014.
- Radio interview, “Villes intelligentes”, *Les années lumière* (ICI Radio-Canada Première), May 2013.
- Print interview, “L’intelligence artificielle au service de l’archéologie”, *Le Devoir*, January 2012.
- Television interview, “Recharge bidirectionnelle de véhicules électriques”, *Campus* (Canal Savoir), March 2011.
- Print interview, “Alimenter le réseau électrique avec sa voiture”, *Impact-Campus*, October 2010.

Awards

- Canadian AI 2020 Best Paper for *Toward adversarial robustness by diversity in an ensemble of specialized deep neural networks*.
- GECCO 2009 Best Paper, *Real-World Application* track, for *Optimizing Low-Discrepancy Sequences with an Evolutionary Algorithm*.
- GECCO 2002 Best Paper, *Evolvable Hardware* track, for *Lens System Design and Re-Engineering with Evolutionary Algorithms*.

Research Grants and Contracts

OPERATING GRANTS OBTAINED AS MAIN APPLICANT

- *Canada-CIFAR Artificial Intelligence Chair*

  CIFAR. $500,000 / 5 years (2025–2030).

- *Neuro-symbolic Approaches for Robust and Interpretable Machine Learning*

Discovery Grant (individual), NSERC. $240,000 / 5 years (2025–2030).

- *Generative Models for Symbolic Models*

  IVADO R^3AI R10 - AI Safety and Alignment. $80,000 / 3 years (2024–2027).

- *Technical team - R8*

  IVADO R^3AI R8 - Healthcare Systems. $75,000 / 3 years (2024–2027).

- *Génération de données synthétiques à la RAMQ*

  Catalyst Grant, CIFAR / Mila. $44,000 / 2 years (2023–2024).

- *L'intelligence artificielle en santé, une formation expérientielle*

  Programme NovaScience: Volet Soutien aux projets, MÉI. $227,393 / 2 years (2021–2023).

- *Canada-CIFAR AI Chair*

  CIFAR. $500,000 / 6 years (2019–2025).

- *Deep Learning with Little Labelled Data*

  Discovery Grant (individual), NSERC. $246,000 / 6 years (2019–2025).

- *DRIFTERS: Deep Radar Interpretation For Tracking and Enhancement of Raw Signal*

  Collaborative Research and Development Grant, NSERC. Partner: Thales Canada. $259,563 / 3 years (2019–2022).

- *Intelligence artificielle appliquée pour l'analyse, l'optimisation et l'innovation*

  Accelerate (cluster of 14 units), Mitacs. Partners: Axes Networks, Can-Explore, Co-Operators, Coveo, Desjardins Assurances Générales, Bentley Canada. $186,667 / 1 year (2019).

- *Assessment of deep learning for analyzing radar signals in maritime environment*

  Accelerate (3 units), Mitacs. Partner: Thales Canada. $45,000 / 1 year (2019).

- *Novel Approaches for Practical Machine Learning*

  PROMPT-Québec. Partners: E Machine Learning and Thales Canada. $365,000 / 2 years (2018–2020).

- *Novel Approaches for Practical Machine Learning*

  Accelerate (cluster of 45 units), Mitacs. Partner: E Machine Learning. $600,000 / 3 years (2016–2020).

- *Personalized Recommendations for a Social Network of Photographers*

  Engage Grant (individual), NSERC. Partner: 500px. $24,926 / 6 months (2016).

- *Sélection de méthodes pour la recommandation personnalisée de documents*

  Engage Grant (individual), NSERC. Partner: Coveo. $24,984 / 6 months (2016).

- *Improving Models for User-Specific State Assessment: A Realtime Querying and Learning Technique*

  Accelerate (1 unit), Mitacs. Partner: Thales Canada. $15,000 / 6 months (2015–2016).

- *Analyse de données massives provenant de médias sociaux*

  Accelerate (1 unit), Mitacs. Partner: Thales Canada. $15,000 / 6 months (2015).

- *Modèle d'évaluation de l'état des transformateurs pour la pérennité et la maintenance*

  Accelerate (2 units), Mitacs. Partner: Hydro-Québec. $30,000 / 6 months (2015).

- *Adaptive Learning Methods for Deeply Embedded Devices*

  Discovery Grant (individual), NSERC. $195,000 / 5 years (2014–2019).

- *Intelligence machine pour la prédiction de l'état de transformateurs de haute tension*

  Engage Grant (individual), NSERC. Partner: Hydro-Québec. $24,992 / 6 months (2014–2015).

- *Smartphone Application for Electric and Conventional Vehicles Data Collection*

  Engage Grant (individual), NSERC. Partner: Thales Canada. $24,987 / 6 months (2014–2015).

- *Installation et essai d'une borne de recharge supportant la technologie "vehicle-to-grid" (V2G)*

  Programme de recherche en partenariat contribuant à la réduction et la séquestration des gaz à effet de serre (team project), FQRNT. $250,000 / 3 years (2010–2013).

- *Integrating Developmental Genetic Programming and Terrain Analysis Techniques in GIS-based Sensor Placement Systems*

  Strategic Industrial Initiative (team project), GEOIDE NCE. $270,000 / 2 years (2010–2012) + $25,000 from MDA Systems Ltd.

- *Enabling Autonomic Computing with Computational Intelligence*

  Discovery Grant (individual), NSERC. $110,000 / 5 years (2009–2014).

- *Apprentissage à grande échelle parallèle pour supercalculateurs*

  New University Researchers Start Up Program (individual), FQRNT. $40,000 / 2 years (2009–2011) + $19,709 for equipment (2009–2010).

- *Fonds de démarrage*

  Université Laval. $10,000 / 1 year (2008–2009).

OPERATING GRANTS OBTAINED AS CO-APPLICANT

- *Advanced Optical Nanoscopy Platform for Neuroscience (AONP)*

  Brain Canada, École Polytechnique de Montréal, The Neuro, and CERVO Research Centre. Co-investigators: Frédéric Leblond, Caroline Ménard, Thomas Stroh, Lucien Weiss, Paul De Koninck, and Flavie Lavoie-Cardinal. $1,050,000 / 3 years (2026–2029).

- *REcherche pour le Développement d'une applIcation de RÉorientation au triage de l'urgence aveC l'inTelligence artificielle (REDIRECT-IA)*

  Project Grant Program, CIHR. Co-investigators: Axel Benhamed, Audrey Durand, Alexis Turgeon, Simon Berthelot, and others. $466,649 / 3 years (2026–2029).

- *Cosmic Insights into Neuronal Connections: AI-Driven Multimodal Synaptic Imaging*

  NOVA, FRQNT and NSERC. Co-investigators: Yves De Koninck, Renée Hložek, Marie-Ève Paquet, Feng Wang, and Flavie Lavoie-Cardinal. $225,000 / 3 years (2025–2028).

- *Projet MAIN: Myo-prothèse à Apprentissage Interactif et Neuro-renforcement*

  Initiative STRATÉGIA, FRQNT. Main applicant: Benoît Gosselin. $886,826 / 3 years (2024–2027).

- *Care trajectory prediction*

  IAR3 R8 – Healthcare Systems, IVADO. Co-investigators: Simon Berthelot, Esli Osmanlliu, and Nadia Lahrichi. $25,000 / 3 years (2024–2027).

- *Union Neurosciences et Intelligence Artificielle Québec (UNIQUE)*

  Strategic Clusters, FRQNT. Main applicant: Karim Jerbi. $3,060,000 / 6 years (2023–2029).

- *Can Astronomy and Machine Learning help detect neurodegeneration?*

  Catalyst Fund, CIFAR. Main applicants: Renée Hložek and Audrey Durand. $50,000 / 2 years (2021–2023).

- *Extreme zooming on intestinal permeability and the western-style diet: Unravelling the role of dietary antigens on the prevalence of cardiometabolic and mental health diseases in the North*

  Deuxième appel à projets majeurs, Sentinelle Nord. Main applicant: Flavie Lavoie-Cardinal and Denis Boudreau. $739,350 / 5 years (2020–2024).

- *Méthodes d'apprentissage automatique pour le développement de la microscopie intelligente des dynamiques cellulaires*

  Projet de recherche en équipe, FRQNT. Main applicant: Flavie Lavoie-Cardinal. $180,000 / 4 years (2020–2023).

- *Déterminer la qualité de la polypharmacie chez les aînés: une approche basée sur l'intelligence artificielle*

Collaborative Health Research Projects, CIHR and NSERC. Main applicant: Caroline Sirois. $1,207,610 / 3 years (2020–2023).

- *Re-penser la découvrabilité, ou comment garantir l'accès à des contenus culturels canadiens dans l'environnement numérique*

  Insight Grants, SSHRC. Main applicant: Véronique Guèvremont. $294,172 / 3 years (2020–2023).

- *Predicting population risk of suicide using health administrative data*

  New Frontiers in Research Fund - Exploration. Main applicant: JianLi Wang. $250,000 / 2 years (2020–2022).

- *REPARTI – Systèmes cyberphysiques et intelligence machine matérialisée*

  Strategic Clusters, FRQNT. Main applicant: Clément Gosselin. $2,880,000 / 6 years (2019–2025).

- *DEpendable and Explainable Learning in Aerospace*

  Collaborative Research and Development Grant, NSERC. Partners: Thales, Bell Helicopter, CAE, Bombardier, CRIAQ. Main applicants: François Laviolette and Giuliano Antoniol. $5,905,510 / 7 years (2019–2026).

- *Machine learning for the insurance industry: predictive models, fraud detection, and fairness*

  Collaborative Research and Development Grant, NSERC. Partner: SSQ Assurance. Main applicant: Mario Marchand. $652,175 / 5 years (2019–2024).

- *Mettre l'IA au service de la diversité des expressions culturelles: une exploration des conditions à remplir pour que les algorithmes de recommandation favorisent la découvrabilité des oeuvres littéraires québécoises dans l'environnement numérique*

  Appel à projets innovants (2019–2022) – Volet 1, OBVIA. Main applicant: Véronique Guèvremont. $159,469 / 3 years (2019–2022).

- *Suivi de la qualité de la pratique de l'électroconvulsivothérapie au Québec basé sur le recueil de données médico-administratives, cliniques et socio-démographiques en contexte réel*

  Données de recherche en contexte réel - Partenariat Innovation-Québec-JANSSEN, FRQS. Main applicant: Alain Lesage. $192,948 / 2 years (2019–2021).

- *Union Neurosciences et Intelligence Artificielle Québec (UNIQUE)*

  Strategic Clusters, FRQNT. Main applicant: Karim Jerbi. $600,000 / 2 years (2019–2021).

- *Big data analytics in insurance*

  Collaborative Research and Development Grant, NSERC. Partner: Intact Financial Corporation. Main applicant: François Laviolette. $2,245,988 / 5 years (2018–2023).

- *PEGASUS-2 - PErsonalized Genomics for prenatal Abnormalities Screening USing maternal blood: Towards First Tier Screening and Beyond*

  Large-scale Applied Research Project Competition, Genome Canada. Main applicants: François Rousseau and Sylvie Langlois. $10,801,250 / 4 years (2018–2022).

- *Nouvelles approches pour le pilotage d'un atelier d'usinage de pièces métalliques de précision basées sur les données*

  Collaborative Research and Development Grant, NSERC. Partner: APN. Main applicant: Jonathan Gaudreault. $230,700 / 5 years (2017–2022).

- *E-Community Health and Toxicity*

  Accelerate (cluster of 118 units), Mitacs. Partners: Two Hat Security. Main applicant: Richard Khoury. $1,693,333 / 3 years (2017–2020).

- *BRITE: Bus RapId Transit systEm*

  Collaborative Research and Development Grant, NSERC. Partners: Thales Canada, LeddarTech. Main applicant: Denis Laurendeau. $426,910 / 3 years (2017–2020).

- *Sécurité urbaine: entraînement, soutien opérationnel, protection des infrastructures et analyses prédictives*

Accelerate (cluster of 42 units), Mitacs. Partners: Thales Canada, UMANX. Main applicant: Sébastien Tremblay. $560,000 / 3 years (2017–2019).

- *Solutions intelligentes pour l'efficience et la fluidité urbaine*

  Accelerate (cluster of 51 units), Mitacs. Partners: Thales Canada, Parc technologique du Québec Métropolitain, and Cascades. Main applicant: Sébastien Tremblay. $680,000 / 3 years (2015–2018).

- *Regroupement stratégique pour l'Étude des Environnements PARTagés Intelligents répartis*

  Strategic Clusters, FRQNT. Main applicant: Denis Laurendeau. $2,150,000 / 6 years (2013–2019).

- *Convergence d'intelligence géospatiale pour l'innovation*

  Appui aux réseaux d'innovation, FRQNT. Main applicant: Mir Abolfazl Mostafavi. $300,000 / 3 years (2013–2016).

- *Simulating cost-effectiveness of screening strategies for preeclampsia risk in pregnant women*

  Operating Grant, CIHR. Main applicants: Daniel Reinharz and Yves Giguère. $91,266 / 2 years (2013–2014).

- *Simulation du coût/efficacité et du coût/utilité du dépistage des gènes de prédisposition au cancer du sein*

  Recherches sur les services de santé, FRQS. Main applicant: François Rousseau. $105,268 / 2 years (2012–2014).

- *Simulating the Cost/Effectiveness of Screening Strategies for Cystic Fibrosis*

  Operating Grant, CIHR. Main applicants: Daniel Reinharz and Patrick Daigneault. $63,815 / 1 year (2011).

- *Infrastructure for Wide Market Adoption of PHEV*

  AUTO21 NCE. Main applicants: Maxime Dubois (2009–2011) and Éric Bibeau (2011–2012). $246,000 / 3 years (2009–2012).

- *NSERC RAAMP – Strategic Network on Robotic Assistants for Automotive Manufacturing Productivity*

  Strategic Network Grant (network establishment), NSERC. Main applicant: Clément Gosselin. $19,600 / 1 year (2009–2010).

- *LSD – Laboratoire de Simulation du Dépistage génétique*

  Operating Grant, CIHR. Main applicant: Daniel Reinharz. $517,233 / 5 years (2008–2013).

- *La simulation comme outil d'évaluation de la pertinence et du retour sur l'investissement des activités en santé publique au Québec*

  Action concertée, FQRSC. Main applicant: Daniel Reinharz. $177,708 / 3 years (2008–2011).

- *Regroupement stratégique pour l'Étude des Environnements PARTagés Intelligents répartis*

  Strategic Cluster, FQRNT. Main applicant: Denis Laurendeau. $2,100,000 / 6 years (2006–2013).

RESEARCH CONTRACTS

- *Intégration de techniques de Programmation Génétique et d'Analyse de Terrain dans un Système de Placement de Capteurs*

  Defence R&D Canada – Valcartier (RDDC Valcartier). $24,900 / 1 year (2012).

- *Development of Multiobjective Optimization Techniques for Sensor Network Layout*

  Defence R&D Canada – Valcartier (RDDC Valcartier). $138,121 / 2 years (2009–2011).

Supervision

PH.D. STUDENTS (ON-GOING)

- **Nizar El Ghazal**, Ph.D. in Electrical Engineering, since 2026
- **Olivier Bussière**, Ph.D. in Computer Science (cosupervisor: Bobin Wang), since 2024 (fast-track to Ph.D. in 2026)

- **Mathis Rezzouk**, Ph.D. in Electrical Engineering (supervisor: Flavie Lavoie-Cardinal), since 2026
- **Arian Yavari**, Ph.D. in Computer Science (supervisor: Jiayi Hong), since 2026
- **Rose Nayoung Kwon**, Ph.D. in Computer Science, since 2025
- **Meryam Chaieb**, Ph.D. in Computer Science (supervisor: Bobin Wang), since 2025
- **Frédéric Beaupré**, Ph.D. in Biophotonics (cosupervisor: Flavie Lavoie-Cardinal), since 2021 (fast-track to Ph.D. in 2022)
- **Jonas Ngnawé**, Ph.D. in Computer Science (cosupervisor: Frédéric Precioso, Université Côte d'Azur, France), since 2022
- **Sara Karami**, Ph.D. in Electrical Engineering, since 2021
- **Sabyasachi Sahoo**, Ph.D. in Electrical Engineering (cosupervisor: Frédéric Precioso, Université Côte d'Azur, France), since 2021
- **Catherine Bouchard**, Ph.D. in Electrical Engineering (cosupervisor: Flavie Lavoie-Cardinal), since 2019 (fast-track to Ph.D. in 2021)
- **Adam Tupper**, Ph.D. in Electrical Engineering, since 2021
- **Benjamin Léger**, Ph.D. in Electrical Engineering, since 2020
- **Sophie Baillargeon**, Ph.D. in Mathematics (specialization in Statistics) (supervisor: Thierry Duchesne), since 2018

MASTER'S STUDENTS WITH THESIS (ON-GOING)

- **Jacob Côté**, M.Sc. in Computer Science (supervisor: Sophie Gobeil), since 2025
- **Camille Godbout**, M.Sc. in Computer Science (supervisor: Bobin Wang), since 2025
- **Katrine Castonguay**, M.Sc. in Computer Science (supervisor: Flavie Lavoie-Cardinal), since 2024

PH.D. STUDENTS (COMPLETED)

- **Fatemeh Nourilenjan Nokabadi**, *Adversarial Robustness of Learning-based Single Object Trackers*, Ph.D. in Electrical Engineering (cosupervisor: Jean-François Lalonde), 2025; now: Data Scientist, Employment and Social Development Canada
- **Arman Afrasiyabi**, *Representation Learning for Few-shot Image Classification*, Ph.D. in Electrical Engineering (cosupervisor: Jean-François Lalonde), 2022; now: Postdoctoral Fellow, Yale University
- **Changjian Shui**, *Principled Deep Learning Approaches for Learning from Limited Labeled Data through Distribution Matching*, Ph.D. in Electrical Engineering (cosupervisor: Boyu Wang, Western Ontario), 2022; now: Postdoctoral Fellow, Vector Institute
- **Mahdieh Abbasi**, *Toward Robust Deep Neural Networks*, Ph.D. in Electrical Engineering (cosupervisor: Denis Laurendeau), 2020; now: Machine Learning Researcher, Mirametrix Inc.
- **Marc-André Gardner**, *Learning to Estimate Indoor Illumination*, Ph.D. in Electrical Engineering (supervisor: Jean-François Lalonde), 2020; now: Applied AI Scientist, Bentley Systems
- **Karol Lina Lopez**, *A Machine Learning Approach for the Smart Charging of Electric Vehicles*, Ph.D. in Electrical Engineering, 2019; now: Data Science Specialist, Ministère des Transports et de la Mobilité durable du Québec
- **Julien-Charles Lévesque**, *Bayesian Hyperparameter Optimization: Overfitting, Ensembles and Conditional Spaces*, Ph.D. in Electrical Engineering (cosupervisor: Robert Sabourin, ÉTS Montréal), 2018; now: Senior Data Scientist, Government of Canada (ESDC)
- **Audrey Durand**, *Déclinaisons de bandits et leurs applications*, Ph.D. in Electrical Engineering (cosupervisor: Joelle Pineau, McGill), 2017; now: Assistant Professor, Université Laval
- **Ahmed Najjar**, *Forage de données de banques administratives en santé*, Ph.D. in Electrical Engineering (cosupervisor: Daniel Reinharz), 2017; now: Data Scientist, Government of Québec (MEES)
- **Vahab Akbarzadeh**, *Spatio-Temporal Coverage Optimization of Sensor Networks*, Ph.D. in Electrical Engineering (cosupervisor: Marc Parizeau), 2016; now: Software Engineer, Arista Networks

- **Zahra Toony**, *Extracting Structured Models From Raw Scans of Manufactured Objects: A Step Towards Embedded Intelligent Handheld 3D Scanning*, Ph.D. in Electrical Engineering (supervisor: Denis Laurendeau), 2015; now: Scientific Developer / Patent Expert, Creaform
- **François-Michel De Rainville**, *Placement interactif de capteurs mobiles dans des environnements tridimensionnels non convexes*, Ph.D. in Electrical Engineering (cosupervisor: Denis Laurendeau), 2015; now: Data Science Specialist, Intact Insurance
- **Meysam Argany**, *Development of a GIS-based method for sensor network deployment and coverage optimization*, Ph.D. in Geomatic Sciences (supervisor: Mir Abolfazl Mostafavi), 2015; now: Assistant Professor, University of Tehran
- **Darwin Brochero**, *Hydroinformatics and Diversity in Hydrological Ensemble Prediction Systems*, Ph.D. in Water Engineering (supervisor: François Anctil), 2013; now: CEO, Arawak

MASTER'S STUDENTS WITH THESIS (COMPLETED)

- **Cynthia García Ybarra**, *Adversarial Random Forest for Synthetic Electronic Health Records Generation*, M.Sc. in Computer Science (cosupervisor: Anne-Sophie Charest), 2025
- **Antoine Séverin Ollier**, *Développement d'un module de détection hyperspectrale et résolu dans le temps pour la microscopie STED*, M.Sc. in Electrical Engineering (supervisor: Flavie Lavoie-Cardinal), 2024
- **Thomas Philippon**, *Robustesse des mécanismes de défense adverse basés sur les ensembles de réseaux de neurones*, M.Sc. in Electrical Engineering, 2023
- **Cyril Blanc**, *Caractérisation automatique d'immeuble depuis une image de façade*, M.Sc. in Electrical Engineering (supervisor: Jean-François Lalonde), 2022
- **Mohamed Abderrahmen Abid**, *Diverse Image Generation with Very Low Resolution Conditioning*, M.Sc. in Electrical Engineering, 2021
- **Gabriel Leclerc**, *Apprendre de données positives et non étiquetées: application à la segmentation et la détection d'événements calciques*, M.Sc. in Electrical Engineering (cosupervisor: Flavie Lavoie-Cardinal), 2021
- **Hugo Siqueira Gomes**, *Meta Learning for Population-Based Algorithms in Black-box Optimization*, M.Sc. in Electrical Engineering, 2021
- **Louis-Émile Robitaille**, *Réseaux de neurones pour l'apprentissage de la préférence en microscopie super-résolution*, M.Sc. in Electrical Engineering (cosupervisors : Audrey Durand and Flavie Lavoie-Cardinal), 2021
- **Sébastien De Blois**, *Deep Learning with Multiple Modalities: Making the Most Out of Available Data*, M.Sc. in Electrical Engineering, 2020
- **El Mehdi Megder**, *Approches basées sur l'apprentissage automatique pour l'anticipation de la qualité d'usinage de pièces métalliques*, M.Sc. in Computer Science (supervisor: Jonathan Gaudreault), 2020
- **Marc-André Gardner**, *Contrôle de la croissance de la taille des individus en programmation génétique*, M.Sc. in Electrical Engineering (cosupervisor: Marc Parizeau), 2014
- **Kevin Tanguy**, *Modélisation et optimisation de la recharge bidirectionnelle de véhicules électriques : application à la régulation électrique d'un complexe immobilier*, M.Sc. in Electrical Engineering (cosupervisor: Maxime Dubois), 2013
- **Audrey Durand**, *Simulation et apprentissage Monte-Carlo de stratégies d'intervention en santé publique*, M.Sc. in Electrical Engineering (cosupervisor: Daniel Reinharz), 2011
- **François-Michel De Rainville**, *Design d'expérimentation interactif : Aide à la compréhension de systèmes complexes*, M.Sc. in Electrical Engineering (supervisor: Denis Laurendeau), 2010

RESEARCH ASSISTANTS

- **Mohamed El Yazid El Yaakoubi**, Software Engineering B.Eng. Student, May to August 2026
- **Samy Mammeri**, Integrated Mathematics - Computer Science B.Sc. Student, May 2025 - now
- **Harold Toukam Zanjio**, Computer Engineering B.Eng. Student, May to August 2021
- **Ruoyu Liu**, Computer Science – Artificial Intelligence M.Sc. Student, May to December 2020

- **Catherine Villeneuve**, Mathematics - Computer Science B.Sc. Student, May to August 2019
- **Keven Voyer**, Computer Science – Artificial Intelligence M.Sc. Student, May to August 2019
- **Philippe-André Luneau**, Mathematics - Computer Science B.Sc. Student, September to December 2018
- **Jonathan Marek**, Computer Engineering B.Eng. Student, May to December 2017
- **Louis-Émile Robitaille**, Computer Engineering B.Eng. Student, May to August 2016
- **Jean-Alexandre Beaumont**, Software Engineering B.Eng. Student, May to August 2016
- **Diane Fournier**, Computer Engineering B.Eng. Student, January 2013 to December 2014
- **Antoine Bois**, Electrical Engineering B.Eng. Student, May 2012 to April 2013
- **Marc-André Gardner**, Computer Engineering B.Eng. Student, May 2009 to April 2012
- **Carl Poirier**, Computer Engineering B.Eng. Student, May 2010 to April 2011
- **Émile Papillon-Corbeil**, Engineering Physics B.Eng. Student, May 2011 to July 2011
- **Camille Besse**, Computer Science Ph.D. Student, June 2010 to August 2010
- **Majid Mallis**, Mathematics - Computer Science B.Sc. Student, January 2009 to December 2009
- **Alexandre Boily**, Computer Engineering B.Eng. Student, May 2009 to August 2010
- **Audrey Durand**, Computer Engineering B.Eng. Student, August 2008 to April 2009

POSTDOCTORAL FELLOWS

- **Fatemeh Gholi Zadeh Kharrat** (cosupervisor: Caroline Sirois), January 2020 to January 2023; now: Researcher, University of California, San Francisco
- **Fan Zhou** (supervisor: Mario Marchand), January to December 2022; now: Assistant Professor, Beihang University
- **Theresa Wiesner** (supervisor: Flavie Lavoie-Cardinal), September 2021 to August 2022; now: Postdoctoral Fellow, Université Aix-Marseille
- **Ihsen Hedhli**, January 2018 to October 2020; now: Research Professional, Université Laval
- **Azadeh Sadat Mozafari**, November 2017 to October 2019; now: Postdoctoral Fellow, York University
- **Farkhondeh Kiaee**, January to December 2016; now: Lecturer, VTU University
- **Matthew Walker**, June 2009 to June 2011; now: Education Technologist, Tech Futures Lab
- **Albert Hung-Ren Ko**, February to October 2010

RESEARCH PROFESSIONALS

- **Mathieu Garon**, IVADO R^3AI R8 - Healthcare system, since May 2025
- **Arman Afrasiyabi**, Computer Vision and Systems Laboratory, October and November 2022
- **Diane Fournier**, Laboratoire de simulation du dépistage génétique, technical supervision, January to August 2015
- **Thierry Moszkowicz**, Computer Vision and Systems Laboratory, supervision at 50%, June 2014 to January 2015
- **Xavier Douville**, Laboratoire de simulation du dépistage génétique, technical supervision, October 2011 to September 2012
- **Sylvain Comtois**, Computer Vision and Systems Laboratory, supervision at 50%, June 2010 to June 2014
- **Julien-Charles Lévesque**, Computer Vision and Systems Laboratory, January to May 2011
- **Mathieu Gagnon**, Laboratoire de simulation du dépistage génétique, technical supervision, September 2009 to August 2011

VISITING INTERNS

- **Lorenzo Dunau**, Master 2 student in Mathematics, Vision and Learning (MVA), ENS Paris-Saclay, France, April to August 2025
- **Soumili Ghosh**, B.Tech. in Computer Science and Engineering, Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, India, May to July 2024
- **Kunal Samanta**, B.Sc. in Mathematics and Computer Science student, Indian Institute of Science (IISc), Bangalore, India, May to July 2023
- **Apoorva Verma**, B.Tech. in Electronics and Communication student, Indian Institute of Technology (IIT), Roorkee, India, May to July 2023
- **Itziar Casado-O'Mara Corral**, Master's in Telecommunication student, University of the Basque Country (UPV/EHU), Spain, May to June 2023
- **Chi Huynh**, Engineering undergraduate student, École Polytechnique, Palaiseau, France, March to August 2023
- **David Wittenberg**, Ph.D. student at the University of Mainz, Germany, April to June 2022
- **Guillaume Camus**, Electronic and Computer Engineering undergraduate student, ENSEA, Cergy-Pontoise, France, November 2019 to February 2020
- **Steeven Janny**, Master 1 student in Electronic, Electric and Automatic, ENS Paris-Saclay, France, May to August 2018
- **Luis Enrique Güitrón**, Computer Engineering undergraduate student, Tecnológico de Monterrey, Santa Fe Campus, Mexico, May to August 2018
- **Sai Krishna Kalyan**, Data Mining and Knowledge Management MSc student, Université Lumière (Lyon 2), France and Universitat Politècnica de Catalunya, Barcelona, Spain, March to August 2017
- **Yosha Tomar**, Electronics and Electrical Engineering undergraduate student, Indian Institute of Technology Guwahati, India, May to July 2017
- **Thibault Parpaite**, Computer Science undergraduate student at University of Bordeaux, France, May to August 2016
- **Farkhondeh Kiaee**, Electrical Engineering Ph.D. student at Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran, May 2014 to November 2015
- **Ludovic Arnold**, Computer Science Ph.D. student at Université Paris-Sud (Paris XI), Orsay, France, March to September 2011
- **Bibhash Kumar Jha**, Mathematics and Computer Science B.Sc. student at Indian Institute of Technology of Kharagpur, India, May to July 2010
- **Juan Luis Jimenez Laredo**, Computer Engineering Ph.D. student at the University of Granada, Spain, October to November 2008

VISITING RESEARCHERS

- **Hamid Boubertakh**, University of Jijel, Algeria, October to November 2010; September to October 2011; May 2012

Publications

- 57 articles in peer-reviewed journals; 85 papers in peer-reviewed conference proceedings; 1 expert and policy reports; 13 technical reports; 20 preprints; 9 other publications without peer review.

ACCEPTED OR PUBLISHED PAPERS IN PEER-REVIEWED SCIENTIFIC JOURNALS

[J57] A. Bilodeau, F. Beaupré, J. Chabbert, K. Thériault, A. Deschênes, J.-M. Bellavance, K. Lessard, R. Bernatchez, P. De Koninck, **C. Gagné**, and F. Lavoie-Cardinal. "A Self-Supervised Foundation Model for Robust and Generalizable Representation Learning in STED Microscopy". *Nature Methods* (2026). In press. bioRxiv: 10.1101/2025.06.06.656993.

[J56] M. Côté, W. Hamadi, C. Laramée, S. Pomerleau, V. Provencher, **C. Gagné**, and B. Lamarche. "Estimating Nutrient Composition of Packaged Foods Using Natural Language Processing and Optimization Modeling". *Current Developments in Nutrition* 10 (July 2026), p. 109387. DOI: 10.1016/j.cdnut.2026.109387.

[J55] J.-F. Pelletier, A. Lesage, **C. Gagné**, F. Gholi Zadeh Kharrat, and J. Wang. "Des systèmes d'aide à la décision clinique dans les salles d'urgence en mobilisant en temps réel, grâce à l'IA, les registres québécois de données de santé quant au risque suicidaire". *Santé mentale au Québec* 51.3 (2026), pp. 295–301. DOI: 10.7202/1129202ar.

[J54] F. Beaupré, A. Bilodeau, T. Wiesner, G. Leclerc, M. Lemieux, G. Nadeau, K. Castonguay, B. Fan, S. Labrecque, R. Hložek, P. De Koninck, **C. Gagné**, and F. Lavoie-Cardinal. "Quantitative Analysis of Miniature Synaptic Calcium Transients Using Positive Unlabeled Deep Learning". *Digital Discovery* 4.1 (2025). DOI: 10.1039/D4DD00197D.

[J53] A. Deschênes, A. Ollier, M. Lafontaine, A. Michaud-Gagnon, J.-G. S. Santiague, A. Bilodeau, **C. Gagné**, P. De Koninck, and F. Lavoie-Cardinal. "Enhancing STED Microscopy via Fluorescence Lifetime Unmixing and filtering in Two-species SPLIT-STED". *Methods in Microscopy* (2025). DOI: 10.1515/mim-2025-0026.

[J52] A. Tupper and **C. Gagné**. "Revisiting Data Augmentation for Ultrasound Images". *Transactions on Machine Learning Research* 7 (2025). URL: https://openreview.net/forum?id=iGcxlTLIL5.

[J51] F. Gholi Zadeh Kharrat, **C. Gagné**, A. Lesage, G. Gariépy, J.-F. Pelletier, C. Brousseau-Paradis, L. Rochette, E. Pelletier, P. Lévesque, M. Mohammed, and J. Wang. "Explainable artificial intelligence models for predicting risk of suicide using health administrative data in Quebec". *PLoS one* 19.4 (2024), e0301117. DOI: 10.1371/journal.pone.0301117.

[J50] J. Li, Y. Lai, R. Wang, C. Shui, S. Sahoo, C. X. Ling, S. Yang, B. Wang, **C. Gagné**, and F. Zhou. "Hessian Aware Low-Rank Perturbation for Order-Robust Continual Learning". *IEEE Transactions on Knowledge and Data Engineering* 36.11 (2024). URL: https://arxiv.org/abs/2311.15161.

[J49] F. Nourilenjan Nokabadi, J.-F. Lalonde, and **C. Gagné**. "Reproducibility Study on Adversarial Attacks Against Robust Transformer Trackers". *Transactions on Machine Learning Research* 5 (2024). URL: https://openreview.net/forum?id=FEEKR0Vl9s.

[J48] J. Wang, F. G. Z. Kharrat, G. Gariépy, **C. Gagné**, J.-F. Pelletier, V. K. Massamba, P. Lévesque, M. Mohammed, and A. Lesage. "Predicting the population risk of suicide using routinely collected health administrative data in Quebec, Canada: Model-Based Synthetic Estimation Study". *JMIR public health and surveillance* 10.1 (2024), e52773. DOI: 10.2196/52773. URL: https://publichealth.jmir.org/2024/1/e52773/.

[J47] C. Bouchard, T. Wiesner, A. Deschênes, A. Bilodeau, B. Turcotte, **C. Gagné**, and F. Lavoie-Cardinal. "Resolution enhancement with a task-assisted GAN to guide optical nanoscopy image analysis and acquisition". *Nature Machine Intelligence* 5 (2023), pp. 830–844. DOI: 10.1038/s42256-023-00689-3.

[J46] S. Lafrenière, F. Gholi-Zadeh-Kharrat, C. Sirois, V. Massamba, L. Rochette, C. Brousseau-Paradis, S. Patry, **C. Gagné**, M. Lemasson, G. Gariépy, C. Mérette, E. Rahme, and A. Lesage. "The 5-year longitudinal diagnostic profile and health services utilization of patients treated with electroconvulsive therapy in Quebec: a population-based study". *Social Psychiatry and Psychiatric Epidemiology* 58.4 (2023), pp. 629–639. DOI: 10.1007/s00127-022-02369-w.

[J45] C. Shui, R. Pu, G. Xu, J. Wen, F. Zhou, **C. Gagné**, C. X. Ling, and B. Wang. "Towards More General Loss and Setting in Unsupervised Domain Adaptation". *IEEE Transactions on Knowledge and Data Engineering* (2023). DOI: 10.1109/TKDE.2023.3266785.

[J44] C. Shui, W. Wang, I. Hedhli, C. M. Wong, F. Wan, B. Wang, and **C. Gagné**. "Lifelong Online Learning from Accumulated Knowledge". *ACM Transactions on Knowledge Discovery from Data* 17.4 (2023). DOI: 10.1145/3563947.

[J43] B. Wang, J. Mendez, C. Shui, F. Zhou, D. Wu, **C. Gagné**, and E. Eaton. "Gap Minimization for Knowledge Sharing and Transfer". *Journal of Machine Learning Research (JMLR)* 24.33 (Jan. 2023). URL: https://jmlr.org/papers/v24/22-0099.html.

[J42] J. Wang, F. G. Z. Kharrat, J.-F. Pelletier, L. Rochette, E. Pelletier, P. Lévesque, V. Massamba, C. Brousseau-Paradis, M. Mohammed, G. Gariépy, **C. Gagné**, and A. Lesage. "A case–control study on predicting population risk of suicide using health administrative data: a research protocol". *BMJ open* 13.2 (2023). DOI: 10.1136/bmjopen-2022-066423.

[J41] D. Wittenberg, F. Rothlauf, and **C. Gagné**. "Denoising autoencoder genetic programming: strategies to control exploration and exploitation in search". *Genetic Programming and Evolvable Machines* 24.2 (2023). DOI: 10.1007/s10710-023-09462-2.

[J40] D. Gourdeau, O. Potvin, P. Archambault, C. Chartrand-Lefebvre, L. Dieumegarde, R. Forghani, **C. Gagné**, A. Hains, D. Hornstein, H. Le, S. Lemieux, M.-H. Lévesque, D. Martin, L. Rosenbloom, A. Tang, F. Vecchio, I. Yang, N. Duchesne, and S. Duchesne. "Tracking and predicting COVID-19 radiological trajectory on chest X-rays using deep learning". *Scientific Reports* 12.1 (2022), p. 5616. DOI: 10.1038/s41598-022-09356-w.

[J39] D. Gourdeau, O. Potvin, J. H. Biem, F. Cloutier, L. Abrougui, P. Archambault, C. Chartrand-Lefebvre, L. Dieumegarde, **C. Gagné**, L. Gagnon, R. Giguère, A. Hains, H. Le, S. Lemieux, M.-H. Lévesque, S. Nepveu, L. Rosenbloom, A. Tang, I. Yang, N. Duchesne, and S. Duchesne. "Deep learning of chest X-rays can predict mechanical ventilation outcome in ICU-admitted COVID-19 patients". *Scientific Reports* 12.1 (2022), p. 6193. DOI: 10.1038/s41598-022-10136-9.

[J38] C. Shui, Q. Chen, J. Wen, F. Zhou, **C. Gagné**, and B. Wang. "A novel domain adaptation theory with Jensen-Shannon divergence". *Knowledge-Based Systems* 257 (Dec. 2022). DOI: 10.1016/j.knosys.2022.109808.

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[J36] S.-C. Kalla, **C. Gagné**, M. Zeng, and L. A. Rusch. "Recurrent neural networks achieving MLSE performance for optical channel equalization." *Optics Express* 29.9 (2021), pp. 13033–13047. DOI: 10.1364/OE.423103.

[J35] C. Sirois, R. Khoury, A. Durand, P.-L. Deziel, O. Bukhtiyarova, Y. Chiu, D. Talbot, A. Bureau, P. Després, **C. Gagné**, F. Laviolette, A.-M. Savard, J. Corbeil, T. Badard, S. Jean, and M. Simard. "Exploring polypharmacy with artificial intelligence: data analysis protocol". *BMC Medical Informatics and Decision Making* 21.1 (2021), pp. 1–8. DOI: 10.1186/s12911-021-01583-x.

[J34] F. Lavoie-Cardinal, A. Bilodeau, M. Lemieux, M.-A. Gardner, T. Wiesner, G. Laramée, **C. Gagné**, and P. De Koninck. "Neuronal activity remodels the F-actin based submembrane lattice in dendrites but not axons of hippocampal neurons". *Scientific reports* 10.1 (2020), pp. 1–17. DOI: 10.1038/s41598-020-68180-2.

[J33] J. Lehman, J. Clune, D. Misevic, C. Adami, L. Altenberg, J. Beaulieu, P. J. Bentley, S. Bernard, G. Beslon, D. M. Bryson, N. Cheney, P. Chrabaszcz, A. Cully, S. Doncieux, F. C. Dyer, K. O. Ellefsen, R. Feldt, S. Fischer, S. Forrest, A. Frénoy, **C. Gagné**, L. Le Goff, L. M. Grabowski, B. Hodjat, F. Hutter, et al. "The surprising creativity of digital evolution: A collection of anecdotes from the evolutionary computation and artificial life research communities". *Artificial Life* 26.2 (2020). DOI: 10.1162/artl_a_00319. URL: https://arxiv.org/abs/1803.03453.

[J32] F. Zhou, C. Shui, M. Abbasi, L.-É. Robitaille, B. Wang, and **C. Gagné**. "Task Similarity Estimation Through Adversarial Multitask Neural Network". *IEEE Transactions on Neural Networks and Learning Systems* 32.2 (2020). DOI: 10.1109/TNNLS.2020.3028022.

[J31] K. L. López, **C. Gagné**, and M.-A. Gardner. "Demand-Side Management using Deep Learning for Smart Charging of Electric Vehicles". *IEEE Transactions on Smart Grid* 10.3 (May 2019). DOI: 10.1109/TSG.2018.2808247.

[J30] A. Durand, T. Wiesner, M.-A. Gardner, L.-É. Robitaille, A. Bilodeau, **C. Gagné**, P. De Koninck, and F. Lavoie-Cardinal. "A machine learning approach for automated optimization of super-resolution optical microscopy". *Nature Communications* 9.5247 (2018). URL: https://www.nature.com/articles/s41467-018-07668-y.

[J29] A. Najjar, D. Reinharz, C. Girouard, and **C. Gagné**. "A Two-Step Approach for Mining Patient Treatment Pathways in Administrative Healthcare Databases". *Artificial Intelligence in Medicine* 87 (May 2018), pp. 34–48. DOI: 10.1016/j.artmed.2018.03.004.

[J28] L. Nshimyumukiza, J.-A. Beaumont, J. Duplantie, S. Langlois, J. Little, F. Audibert, C. McCabe, J. Gekas, Y. Giguère, C. Gagné, D. Reinharz, and F. Rousseau. "Cell-Free DNA–Based Non-invasive Prenatal Screening for Common Aneuploidies in a Canadian Province: A Cost-Effectiveness Analysis". *Journal of Obstetrics and Gynaecology Canada* 40.1 (Jan. 2018), pp. 48–60. DOI: 10.1016/j.jogc.2017.05.015.

[J27] M.-A. Gardner, K. Sunkavalli, E. Yumer, X. Shen, E. Gambaretto, **C. Gagné**, and J.-F. Lalonde. "Learning to Predict Indoor Illumination from a Single Image". *ACM Transactions on Graphics (SIGGRAPH Asia)* 9.4 (Nov. 2017). URL: https://arxiv.org/abs/1704.00090.

[J26] F. Kiaee, **C. Gagné**, and H. Sheikhzadeh. "A Double-Layer ELM with Added Feature Selection Ability using a Sparse Bayesian Approach". *Neurocomputing* 216 (Dec. 2016), pp. 371–380. DOI: 10.1016/j.neucom.2016.08.011.

[J25] L. Nshimyumukiza, X. Douville, D. Fournier, J. Duplantie, R. Daher, I. Charlebois, J. Longtin, J. Papenburg, M. Guay, M. Boissinot, M. G. Bergeron, D. Boudreau, **C. Gagné**, F. Rousseau, and D. Reinharz. "Cost effectiveness analysis of antiviral treatment in the management of seasonal influenza A: point-of-care rapid test versus clinical judgment". *Influenza and Other Respiratory Viruses* 10.2 (Mar. 2016), pp. 113–121. DOI: 10.1111/irv.12359.

[J24] K. Tanguy, M. Dubois, K. L. Lopez, and **C. Gagné**. "Optimization Model and Economic Assessment of Collaborative Charging using Vehicle-To-Building". *Sustainable Cities and Society* 26 (Oct. 2016), pp. 496–506. DOI: 10.1016/j.scs.2016.03.012.

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[J22] M.-A. Gardner, **C. Gagné**, and M. Parizeau. "Controlling Code Growth by Dynamically Shaping the Genotype Size Distribution". *Genetic Programming and Evolvable Machines* 16.4 (2015), pp. 455–498. DOI: 10.1007/s10710-015-9242-8.

[J21] K. L. Lopez, **C. Gagné**, G. Castellanos-Dominguez, and M. Orozco-Alzate. "Training subset selection in Hourly Ontario Energy Price forecasting using time series clustering-based stratification". *Neurocomputing* 156.25-05-2015 (2015), pp. 268–279. DOI: 10.1016/j.neucom.2014.12.052.

[J20] Z. Toony, D. Laurendeau, and **C. Gagné**. "Describing 3D Geometric Primitives Using the Gaussian Sphere and the Gaussian Accumulator". *3D Research* 6.4 (Dec. 2015). DOI: 10.1007/s13319-015-0074-3.

[J19] V. Akbarzadeh, J.-C. Lévesque, **C. Gagné**, and M. Parizeau. "Efficient Sensor Placement Optimization Using Gradient Descent and Probabilistic Coverage". *Sensors* 14 (2014), pp. 15525–15552. DOI: 10.3390/s140815525.

[J18] L. Nshimyumukiza, A. Bois, P. Daigneault, L. Lands, A.-M. Laberge, D. Fournier, J. Duplantie, Y. Giguère, J. Gekas, **C. Gagné**, F. Rousseau, and D. Reinharz. "Cost-Effectiveness of Newborn Screening for Cystic Fibrosis: A Simulation Study". *Journal of Cystic Fibrosis* 13.3 (2014), pp. 267–274. DOI: 10.1016/j.jcf.2013.10.012.

[J17] V. Akbarzadeh, C. Gagné, M. Parizeau, M. Argany, and M. A. Mostafavi. "Probabilistic Sensing Model for Line-of-sight Coverage-based Sensor Placement Optimization". *IEEE Transactions on Instrumentation and Measurement* 62.2 (Feb. 2013), pp. 293–303. DOI: 10.1109/TIM.2012.2214952.

[J16] J. Duplantie, O. M. Gonzalez, A. Bois, L. Nshimyumukiza, J. Gekas, E. Bujold, V. Morin, M. Vallée, Y. Giguère, **C. Gagné**, F. Rousseau, and D. Reinharz. "Cost-Effectiveness of the Management of Rh-Negative Pregnant Women". *Journal of Obstetrics and Gyneacology of Canada* 35.8 (2013), pp. 730–740.

[J15] L. Nshimyumukiza, A. Durand, M. Gagnon, X. Douville, S. Morin, C. Lindsay, J. Duplantie, C. Gagné, S. Jean, Y. Giguère, S. Dodin, F. Rousseau, and D. Reinharz. "An economic evaluation: Simulation of the cost/effectiveness and cost/utility of universal prevention strategies against osteoporosis-related fractures". *Journal of Bone and Mineral Research* 28.2 (2013), pp. 383–394.

[J14] L. Nshimyumukiza, J. Duplantie, M. Gagnon, X. Douville, D. Fournier, C. Lindsay, M. Parent, A. Milot, Y. Giguère, **C. Gagné**, F. Rousseau, and D. Reinharz. "Dabigatran versus warfarin under standard or pharmacogenetic-guided management for the prevention of stroke and systemic thromboembolism in patients with atrial fibrillation: a cost/utility analysis using an analytic decision model". *Thrombosis Journal* 11.14 (2013).

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[J10] A. Durand, C. Gagné, L. Nshimyumukiza, M. Gagnon, F. Rousseau, Y. Giguère, and D. Reinharz. "Population-based Simulation for Public Health: Generic Software Infrastructure and its Application to Osteoporosis". *IEEE transactions on Systems, Man, and Cybernetics, Part A* 42.6 (2012), pp. 1396–1409.

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[J7] D. Brochero, F. Anctil, and C. Gagné. "Simplifying a Hydrological Ensemble Prediction System with a Backward Greedy Selection of Members, Part II: Generalization in Time and Space". *Hydrology and Earth System Sciences* 15.11 (2011), pp. 3327–3341.

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[T6] C. Gagné. *Classification and Case-Studies of Pursuit-Evasion Problems*. Contract report. Québec City (QC), Canada: Informatique WGZ Inc., June 2007.

[T5] C. Gagné. *Experiments with a Simple Scenario for Model-Checking Pursuit-Evasion Problems*. Contract report. Québec City (QC), Canada: Informatique WGZ Inc., June 2007.

[T4] C. Gagné. *PEGGI: A Tool to Generate Specifications for Model-Checking Pursuit-Evasion Problems*. Contract report. Québec City (QC), Canada: Informatique WGZ Inc., June 2007.

[T3] C. Gagné and C. Liu. *Analysis and Synthesis of Protocols for Pursuit-Evasion Problems*. Contract report. Québec City (QC), Canada: Informatique WGZ Inc., Oct. 2007.

[T2] C. Gagné. *Open BEAGLE Compilation HOWTO*. Tech. rep. RT-LVSN-2003-02-V301-R. Laboratoire de vision et systèmes numériques, Université Laval, Oct. 2005.

[T1] C. Gagné and M. Parizeau. *Open BEAGLE Manual*. Tech. rep. RT-LVSN-2003-01-V300-R1. Laboratoire de vision et systèmes numériques, Université Laval, Oct. 2005.

PREPRINTS

[P20] B. Fan, A. Bilodeau, F. Beaupré, T. Wiesner, **C. Gagné**, F. Lavoie-Cardinal, and R. Hložek. "Ca2+ transient detection and segmentation with the Astronomically motivated algorithm for Background Estimation And Transient Segmentation (Astro-BEATS)". *ArXiv e-prints* 2603.22311 (Mar. 2026). URL: https://arxiv.org/abs/2603.22311.

[P19] B. Léger, S. Gupta, S. Mammeri, K. Meidani, C. Subakan, and **C. Gagné**. "SNIP++: Fine-Grained Symbolic-Numerical Alignment for Symbolic Regression". *ArXiv e-prints* 2609.31965 (Sept. 2026). URL: https://arxiv.org/abs/2609.31965.

[P18] R. Luo, **C. Gagné**, J. Ngnawé, I. Ullah, and K. Morrissey. "A Stochastic–Geometric Theory of Scaling Laws in Grokking". *ArXiv e-prints* 2606.30388 (June 2026). URL: https://arxiv.org/abs/2606.30388.

[P17] J. Ngnawé, Y. Pequignot, S. Sahoo, **C. Gagné**, F. Precioso, and S. Koyejo. "Mind the Spike: Mechanisms and Brittleness of Visual Massive Activations in Large Vision-Language Models". *ArXiv e-prints* 2609.32808 (Sept. 2026). URL: https://arxiv.org/abs/2609.32808.

[P16] V. Singh, D. Ganguly, W. Chen, S. Sahoo, S. Sankar, B. Zhang, M. Hariri, S. Wang, O. Zafar, **C. Gagné**, and V. Chaudhary. "Reliability-Gated Source Anchoring for Continual Test-Time Adaptation". *ArXiv e-prints* 2605.14063 (May 2026). URL: https://arxiv.org/abs/2605.14063.

[P15] A. Tupper and **C. Gagné**. "Analyzing Data Augmentation for Medical Images: A Case Study in Ultrasound Images". *ArXiv e-prints* 2403.09828 (Mar. 2024). URL: https://arxiv.org/abs/2403.09828.

[P14] M. Abid, A. Afrasiyabi, I. Hedhli, J.-F. Lalonde, and **C. Gagné**. "Domain Agnostic Image-to-image Translation using Low-Resolution Conditioning". *ArXiv e-prints* 2305.05023 (May 2023). URL: https://arxiv.org/abs/2305.05023.

[P13] C. Bouchard, V. Boulanger, F. Lavoie-Cardinal, and **C. Gagné**. "Filtering Pixel Latent Variables for Unmixing Volumetric Images". *ArXiv e-prints* 2312.05357 (Dec. 2023). URL: https://arxiv.org/abs/2312.05357.

[P12] T. Philippon and **C. Gagné**. "Improved Robustness Against Adaptive Attacks With Ensembles and Error-Correcting Output Codes". *ArXiv e-prints* 2303.02322 (Mar. 2023). URL: https://arxiv.org/abs/2303.02322.

[P11] W. W. Wang, G. Xu, R. Pu, J. Li, F. Zhou, C. Shui, C. Ling, **C. Gagné**, and B. Wang. "Evolving Domain Generalization". *ArXiv e-prints* 2206.00047 (June 2022). URL: https://arxiv.org/abs/2206.00047.

[P10] M. A. Abid, I. Hedhli, and **C. Gagné**. "A Generative Model for Hallucinating Diverse Versions of Super Resolution Images". *ArXiv e-prints* 2102.06624 (Feb. 2021). URL: https://arxiv.org/abs/2102.06624.

[P9] M. A. Abid, I. Hedhli, J.-F. Lalonde, and **C. Gagné**. "Image-to-Image Translation with Low Resolution Conditioning". *ArXiv e-prints* 2107.11262 (July 2021). URL: https://arxiv.org/abs/2107.11262.

[P8] H. S. Gomes, B. Léger, and **C. Gagné**. "Meta Learning Black-Box Population-Based Optimizers". *ArXiv e-prints* 2103.03526 (Mar. 2021). URL: https://arxiv.org/abs/2103.03526.

[P7] M. Abbasi, D. Laurendeau, and **C. Gagné**. "Self-supervised Robust Object Detectors from Partially Labelled datasets". *ArXiv e-prints* 2005.11549 (May 2020). URL: https://arxiv.org/abs/2005.11549.

[P6] A. S. Mozafari, H. S. Gomes, and **C. Gagné**. "A Novel Unsupervised Post-Processing Calibration Method for DNNS with Robustness to Domain Shift". *ArXiv e-prints* 1911.11195 (Nov. 2019). URL: https://arxiv.org/abs/1911.11195.

[P5] M. Abbasi, A. Rajabi, A. Mozafari, R. B. Bobba, and **C. Gagné**. "Controlling Over-generalization and its Effect on Adversarial Examples Generation and Detection". *ArXiv e-prints* 1808.08282 (Aug. 2018). URL: https://arxiv.org/abs/1808.08282.

[P4] A. Cervantes, **C. Gagné**, P. Isasi, and M. Parizeau. "Evaluating and Characterizing Incremental Learning from Non-Stationary Data". *ArXiv e-prints* 1806.06610 (June 2018). URL: https://arxiv.org/abs/1806.06610.

[P3] A. S. Mozafari, L. W. Siqueira Gomes Hugo, S. Janny, and **C. Gagné**. "Attended Temperature Scaling: A Practical Approach for Calibrating Deep Neural Networks". *ArXiv e-prints* 1810.11586 (Oct. 2018). URL: https://arxiv.org/abs/1810.11586.

[P2] A. Durand and **C. Gagné**. "Estimating Quality in User-Guided Multi-Objective Bandits Optimization". *ArXiv e-prints* 1701.01095 (Jan. 2017). URL: https://arxiv.org/abs/1701.01095.

[P1] F. Kiaee, **C. Gagné**, and M. Abbasi. "Alternating Direction Method of Multipliers for Sparse Convolutional Neural Networks". *ArXiv e-prints* 1611.01590 (Nov. 2016). URL: https://arxiv.org/abs/1611.01590.

PUBLICATIONS WITHOUT PEER REVIEW

[O9] A. Najjar, **C. Gagné**, and D. Reinharz. "Patient Treatment Pathways Clustering". *NIPS 2015 Workshop on Machine Learning in Healthcare*. 2015. URL: http://vision.gel.ulaval.ca/~cgagne/pubs/mlhc-nips2015.pdf.

[O8] F.-M. D. Rainville, F.-A. Fortin, M.-A. Gardner, M. Parizeau, and **C. Gagné**. "DEAP – Enabling Nimbler Evolutions". *SIGEVOlution* 6.2 (Feb. 2014), pp. 17–26. DOI: 10.1145/2597453.2597455.

[O7] D. Brochero, F. Anctil, **C. Gagné**, and K. L. Lopez. "Finding Diversity for Building One-day Ahead Hydrological Ensemble Prediction System based on Artificial Neural Network Stacks". *European Geosciences Union (EGU), Geophysical Research Abstract*. Vol. 15. Apr. 2013.

[O6] D. Brochero, F. Anctil, and C. Gagné. "Comparison of three methods for the optimal allocation of hydrological model participation in an Ensemble Prediction System". *European Geosciences Union (EGU) General Assembly 2012, Geophysical Research Abstract*. 2012.

[O5] D. Brochero, F. Anctil, and C. Gagné. "Forward Greedy ANN input selection in a stacked framework with Adaboost.RT - A streamflow forecasting case study exploiting radar rainfall estimates". *European Geosciences Union (EGU) General Assembly 2012, Geophysical Research Abstract*. 2012.

[O4] F. Anctil, D. Brochero, and C. Gagné. "Which Optimization Criterion Leads to the Reliable Simplification of a Hydrological Ensemble Prediction System with a Backward Greedy Selection of Members?" *European Geosciences Union (EGU) General Assembly 2011, Geophysical Research Abstracts*. 2011.

[O3] C. Gagné and M. Parizeau. "Open BEAGLE, A C++ Framework for your Favorite Evolutionary Algorithm". *SIGEVOlution* 1.1 (2006), pp. 12–14.

[O2] C. Gagné, M. Parizeau, and M. Dubreuil. "A Robust Master-Slave Distribution Architecture for Evolutionary Computations". *Late Breaking Papers at GECCO*. 2003.

[O1] C. Gagné and M. Parizeau. "Open BEAGLE: A New Versatile C++ Framework for Evolutionary Computations". *Late Breaking Papers at GECCO*. 2002.

Software

- *DEAP: Distributed Evolutionary Algorithms in Python*. Open source software available at `https://github.com/deap/deap`.
- *SCHNAPS: Generic Population-based Simulator for Public Health*. Open source software available at `https://github.com/audurand/schnaps`.
- *Open BEAGLE: A Generic C++ Evolutionary Computation Framework*. Open source software available at `https://github.com/chgagne/beagle`.
- *BEAGLE Puppy: A Minimalist GP Library in C++*. Open source software available at `http://beagle.gel.ulaval.ca/puppy`.

Patents

- *Method and system for dynamically estimating light-source parameters from multiple images*. US Patent No. 18,009,721, issued 2024 (filed 2023). Inventors: Marc-André Gardner, Jean-François Lalonde, and Christian Gagné.
- *Generating light-source-specific parameters for digital images using a neural network*. US Patent No. 18,062,460, issued 2024 (filed 2022). Inventors: Kalyan Sunkavalli, Yannick Hold-Geoffroy, Christian Gagné, Marc-André Gardner, and Jean-François Lalonde.
- *Dynamically estimating light-source-specific parameters for digital images using a neural network*. US Patent No. 11,538,216, issued 2022 (filed 2019). Inventors: Kalyan Sunkavalli, Yannick Hold-Geoffroy, Christian Gagné, Marc-André Gardner, and Jean-François Lalonde.

Teaching

- GIF-7010 – Advances in Machine Learning, 2024–2026.
- GIF-4101/GIF-7005/GIF-7015 – Introduction to Machine Learning, 2009–2026.
- GIF-3004 – Real-Time Embedded Systems, 2017–2022.
- GIF-3000 – Computer Architecture, 2010–2016.
- GEL-3005 – Design IV (synthesis), 2008–2014.
- GEL-1001 – Design I (methodology), 2009–2011.
- IFT-19968 – Algorithms for the Engineer II, 2009.
- IFT-18254 – Distributed Computer Systems, 2005.

Invited Talks and Presentations

- Keynote, *Intelligence artificielle et recherche : où en sommes-nous et que pouvons-nous en faire ?*, CR-CHUL, Journée de l'axe Maladies infectueuses et immunitaires, Quebec City, Canada, 2026.
- Keynote, *Souveraineté numérique et IA : quelle trajectoire pour le Québec?*, Rendez-vous IA - Humain, Rouyn-Noranda, Canada, 2026.
- Keynote, *Souveraineté numérique et IA : quelle trajectoire pour le Québec?*, Webinaires de l'IID, Quebec City, Canada, 2026.
- Invited talk, *The Case for National AIs*, Shaping the Future of Artificial Intelligence and Regulation, Montreal, Canada, 2025.
- Invited talk, *Plaidoyer pour une IA nationale*, Rendez-vous IA Québec, Quebec City, Canada, 2025.
- Invited talk, *L'avenir et la qualité de la recherche scientifique à l'ère de l'intelligence artificielle (IA)*, École doctorale du CRIMT, Quebec City, Canada, 2024.
- Keynote, *IA responsable: de quoi parle-t-on au juste?*, Atelier sur une IA responsable en entreprise, organisé par Confiance IA, Quebec City, Canada, 2024.
- Keynote, *Intelligence artificielle et recherche : où en sommes-nous et que pouvons-nous en faire ?*, 16e journée de la recherche du Groupe de recherche en écologie buccale, Quebec City, Canada, 2024.
- Invited talk, *Ingrédients humains : connaissances et compétences*, Rencontre des administratrices et administrateurs 2024 – Université Laval, Quebec City, Canada, 2024.
- Invited talk, *Intelligence artificielle et recherche : où en sommes-nous et que pouvons-nous en faire ?*, Retraite du département de biochimie, microbiologie et bio-informatique, Quebec City, Canada, 2024.
- Invited talk, *Le potentiel transformatif de l'IA en recherche ?*, Atelier: L'impact de l'IA sur les sciences humaines et sociales, organisé par OBVIA, Quebec City, Canada, 2024.
- Invited talk, *Où en sommes-nous avec l'IA?*, Instance de concertation sur l'IA en enseignement supérieur, Ministère de l'enseignement supérieur, Quebec City, Canada, 2024.
- Invited talk, *Prenez le virage de l'intelligence artificielle*, Séminaire aux étudiants du Collège Montmorency, Laval, Canada, 2023.
- Invited talk, *Artificial Intelligence for Critical Systems*, Meeting with ICAO / MRIF Québec, in the presence of Mr. Juan Carlos Salazar, secretary-general of ICAO, Quebec City, Canada, 2023.
- Invited talk, *Prenez le virage de l'intelligence artificielle*, Journée périscolaire – Cégep de Ste-Foy, Quebec City, Canada, 2023.
- Invited talk, *ART-ificiel : création littéraire et intelligence artificielle*, À carnets ouverts - Maison de la Littérature, Quebec City, Canada, 2022.
- Invited talk, *Apprentissage automatique et apprentissage profond : domaines phares de l'intelligence artificielle moderne*, Séminaire du LI2A, UQTR, Trois-Rivière, Canada, 2022.
- Invited talk, *Peut-on faire de l'IA avec peu de données?*, TimeWorld IA 2022 - Montréal, Montreal, Canada, 2022.
- Invited talk, *Artificial Intelligence and Data Science at Université Laval*, Brasilian economic delegation, organized by the Ministry of Foreign Affairs, Canada, Quebec City, Canada, 2022.
- Invited talk, *La recherche comme levier pour réussir sa transformation vers l'IA : de la R&D à l'IID*, Webinaires de l'IID, Quebec City, Canada, 2021.
- Invited talk, *Research on AI and data in Quebec City*, The State of AI Research in Quebec and Mexico, Quebec City (online), Canada, 2021.
- Invited talk, *Efficient Learning and Optimization from Sparse Feedback*, Journée du CeRVIM, Quebec City, Canada, 2021.
- Invited talk, *Intelligence artificielle, possibilités et réalité*, École d'été de l'Institut du patrimoine culturel (IPAC), Quebec City, Canada, 2021.

- Presentation, *Toward Adversarial Robustness by Diversity in an Ensemble of Specialized Deep Neural Networks*, Canadian AI, Ottawa (virtual), Canada, 2020.
- Invited talk, *Meta-learning of optimization techniques*, Benchmarked: Optimization meets Machine Learning, Online, Netherlands, 2020.
- Invited talk, *Perspectives sur l'intelligence artificielle*, ISACA Québec 2019, congrès international sur les opportunités et les défis des technologies émergentes, Canada, 2019.
- Invited talk, *L'écosystème IA de Québec 2019*, Rendez-vous IA Québec 2019, Canada, 2019.
- Keynote, *Utliisation des données massives dans les opérations et ingénierie des systèmes intelligents*, Colloque du sous-ministériat à l'ingénierie et aux infrastructures 2019 du Ministère des transports du Québec, Québec, Canada, 2019.
- Keynote, *Le potentiel et les enjeux de l'intelligence artificielle*, Conférence annuelle TELUS Santé, Québec, Canada, 2018.
- Invited talk, *Portrait de l'écosystème en intelligence artificielle à Québec*, Rendez-vous IA de Québec 2018, Québec, Canada, 2018.
- Keynote, *Prenez le virage de l'intelligence artificielle*, Retrouvailles Université Laval, Québec, Canada, 2018.
- Presentation, *Deep Learning Methods for Smart Charging of Electric Vehicles*, Montreal AI Symposium, Montréal, Canada, 2018.
- Invited talk, *Recharge intelligente de véhicules électriques*, Symposium 3i sur le réseau intelligent, Montréal, Canada, 2017.
- Invited talk, *Des systèmes intelligents avec humains dans la boucle*, Objectifs numériques, Québec, Canada, 2017.
- Invited talk, *Recharge intelligente et bidirectionnelle de véhicules électriques*, Intelligence numérique / Digital Intelligence 2016 (#DI2016), Québec, QC, Canada, 2016.
- Keynote, *Développer le potentiel des véhicules électriques par la recharge intelligente bidirectionnelle*, Colloque « Présent et avenir des transports électrifiés, intelligents et interconnectés au Québec », Institut des technologies de l'information et société, Université Laval, Québec, Canada, 2014.
- Presentation, *Tutorial on Evolutionary Computation for Supervised Learning*, ACM SIGEVO Genetic and Evolutionary Computation Conference (GECCO 2013), Amsterdam, Netherlands, 2013.
- Invited talk, *Réseaux de capteurs et transports intelligents : une vision d'avenir*, Colloque international sur les villes intelligentes, Québec, Canada, 2013.

Scientific Events Organization

- Co-organizer, *Rendez-vous IA Québec*, Québec, QC, 2018–2023.
- Co-organizer, *First Workshop on Interactive Labeling and Data Augmentation for Vision*, ICCV 2021.
- Publicity chair, *Genetic and Evolutionary Computation Conference* (GECCO) 2014, Vancouver, BC.
- Responsible, Section 200 (Physical Sciences, Mathematics, and Engineering), Scientific Committee of the 80th Congress of the Acfas (French-speaking Association for the Advancement of Knowledge), Montreal, QC, 2012.
- Co-organizer, *Evolutionary Art Competition*, GECCO 2009–2012.
- Organizer, *Undergraduate Student Workshop*, GECCO 2011, Dublin, Ireland.
- Competitions chair, GECCO 2010, Portland, OR, 2010.
- Local chair, GECCO 2009, Montreal, QC, 2009.
- Sponsors chair, *High Performance Computing Symposium* (HPCS), Québec, QC, 2008.

Committees

INTERNATIONAL COMMITTEE

- Scientific Committee, DIM AI4IDF, since 2023.
- Executive Board, ACM Special Interest Group on Evolutionary Computation (SIGEVO), 2017–2023.

NATIONAL COMMITTEE

- National Resources Allocation Committee, Compute Canada (now the Digital Research Alliance of Canada), 2009–2013, 2017.

JOURNAL EDITORIAL BOARDS

- Editorial Committee, *Genetic Programming and Evolvable Machines*, since 2013.
- Guest Editor, *International Journal of Arts and Technology (IJART)*, special section, 2015.

CONFERENCE PROGRAM COMMITTEES

- Area chair, *Neural Information Processing Systems* (NeurIPS), 2026.
- Track co-chair, *Neuroevolution* track, *Genetic and Evolutionary Computation Conference* (GECCO), 2026 and 2027.
- Track co-chair, *Evolutionary Machine Learning* track, *Genetic and Evolutionary Computation Conference* (GECCO), 2021.
- Track co-chair, *Digital Entertainment Technology and Art* track, *Genetic and Evolutionary Computation Conference* (GECCO), 2011.

LOCAL COMMITTEES

- Advising committee on artificial intelligence, Université Laval, since 2023.
- Graduate programs committee, Computer Science and Software Engineering Department, 2018–2021.
- Graduate programs committee, Electrical Engineering and Computer Engineering Department, 2017–2021.
- Respondent for Computer Engineering, table des répondants pour l'accréditation des programmes de génie, Université Laval, 2017–2018.
- Undergraduate programs committee, Electrical Engineering and Computer Engineering Department, 2010–2018.
- Working group on transportation electrification, Institut Technologies de l'information et société, 2014.
- Secretary of the assembly, Electrical Engineering and Computer Engineering Department, 2008–2011.
- President of the working group on microprocessor teaching, undergraduate programs committee, Electrical Engineering and Computer Engineering Department, 2008–2009.

Professional Associations

- Ordre des ingénieurs du Québec (OIQ), Engineer-in-Training (E.I.T.) between 2000 and 2011, Engineer (Eng.) since 2011.
- Institute of Electrical and Electronics Engineers (IEEE), member since 2008.

Miscellaneous

- Languages: French (native), English (excellent).

- Citizenship: Canadian.

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